

JEE MAINS GRAND TEST SERIES



NARAYANA
IIT ACADEMY
INDIA

40+
YEARS
OF EXCELLENCE

Sec: SR.IIT_*CO-SC(MODEL-A,B&C)
Time: 3HRS

Date:14-01-24
Max. Marks: 300

Name of the Student: _____

H.T. NO:

14-01-24_SR.STAR CO-SUPER CHAINA(MODEL-A,B&C)_JEE-MAIN_GTM-16(N)_SYLLABUS

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

MATHEMATICS: **TOTAL SYLLABUS**

SUBJECT	MISTAKES		
	JEE SYLLABUS Q'S	JEE EXTRA SYLLABUS Q'S	TOTAL Q'S
MATHS			
PHYISCS			
CHEMISTRY			

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SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. Identify the pair of physical quantities that have same dimensions :
 - 1) velocity gradient and decay constant
 - 2) wien's constant and Stefan constant
 - 3) angular frequency and angular momentum
 - 4) wave number and Avogadro number
2. The distance between Sun and Earth is R . The duration of year if the distance between Sun and Earth becomes $3R$ will be :
 - 1) $\sqrt{3}$ years
 - 2) 3 years
 - 3) 9 years
 - 4) $3\sqrt{3}$ years
3. A stone of mass m , tied to a string is being whirled in a vertical circle with a uniform speed. The tension in the string is :
 - 1) the same throughout the motion
 - 2) minimum at the highest position of the circular path
 - 3) minimum at the lowest position of the circular path
 - 4) minimum when the rope is in the horizontal position
4. Two identical charged particles each having a mass 10 g and charge $2.0 \times 10^{-7} \text{ C}$ are placed on a horizontal table with a separation of L between them such that they stay in limited equilibrium. If the coefficient of friction between each particle and the table is 0.25, find the value of L . [Use $g = 10 \text{ ms}^{-2}$]
 - 1) 12 cm
 - 2) 10 cm
 - 3) 8 cm
 - 4) 5 cm
5. Magnification produced by astronomical telescope for normal adjustment is 10 and length of telescope is 1.1m. The magnification when the image is formed at least distance of distinct vision ($D = 25 \text{ cm}$) is
 - 1) 14
 - 2) 6
 - 3) 16
 - 4) 18

6. Two massless springs with spring constants $2k$ and $2k$, carry 50 g and 100 g masses at their free ends. These two masses oscillate vertically such that their maximum velocities are equal. Then, the ratio of their respective amplitudes will be

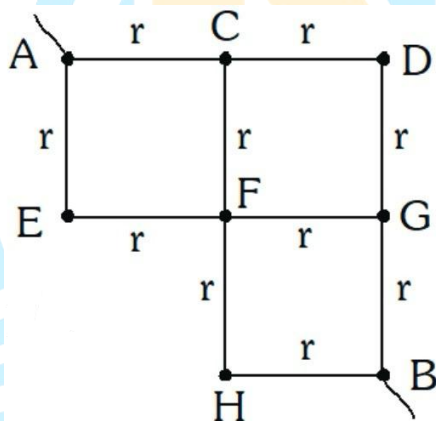
1) $1 : 2$ 2) $3 : 2$ 3) $3 : 1$ 4) $2 : 3$

7. What will be the most suitable combination of three resistors

$A = 2\Omega$, $B = 4\Omega$, $C = 6\Omega$ so that $\left(\frac{22}{3}\right)\Omega$ is equivalent resistance of combination?

- 1) Parallel combination of A and C connected in series with B.
- 2) Parallel combination of A and B connected in series with C.
- 3) Series combination of A and C connected in parallel with B.
- 4) Series combination of B and C connected in parallel with A.

8. Find equivalent resistance R between points A and B for the circuit shown in drawing if resistance of each link is as shown



1) $\frac{3r}{4}$ 2) $\frac{4r}{3}$ 3) $\frac{12r}{7}$ 4) $\frac{7r}{12}$

9. A proton, a deuteron and an α -particle with same kinetic energy enter into a uniform magnetic field at right angle to magnetic field. The ratio of the radii of their respective circular paths is :

1) $1 : \sqrt{2} : \sqrt{2}$ 2) $1 : 1 : \sqrt{2}$ 3) $\sqrt{2} : 1 : 1$ 4) $1 : \sqrt{2} : 1$

10. Given below are two statements :

Statement-I : The reactance of an ac circuit is zero. It is possible that the circuit contains a capacitor and an inductor.

Statement-II : In ac circuit, the average power delivered by the source never becomes zero.

In the light of the above statements, choose the correct answer from the options given below :

1) Both Statement I and Statement II are true.

2) Both Statement I and Statement II are false.

3) Statement I is true but Statement II is false.

4) Statement I is false but Statement II is true.

11. potential energy as a function of r is given by $U = \frac{A}{r^{10}} - \frac{B}{r^5}$, where r is the interatomic distance, A and B are positive constants. The equilibrium distance between the two atoms will be.

1) $\left(\frac{A}{B}\right)^{\frac{1}{5}}$

2) $\left(\frac{B}{A}\right)^{\frac{1}{5}}$

3) $\left(\frac{2A}{B}\right)^{\frac{1}{5}}$

4) $\left(\frac{B}{2A}\right)^{\frac{1}{5}}$

12. An object of mass 5 kg is thrown vertically upwards from the ground. The air resistance produces a constant retarding force of 10 N throughout the motion. The ratio of time of ascent to the time of descent will be equal to : $[Use\ g = 10\ ms^{-2}]$

1) 1 : 1

2) $\sqrt{2} : \sqrt{3}$

3) $\sqrt{3} : \sqrt{2}$

4) 2 : 3

13. A fly wheel is accelerated uniformly from rest and rotates through 5 rad in the first second. The angle rotated by the fly wheel in the next second, will be :

1) 7.5 rad

2) 15 rad

3) 20 rad

4) 30 rad

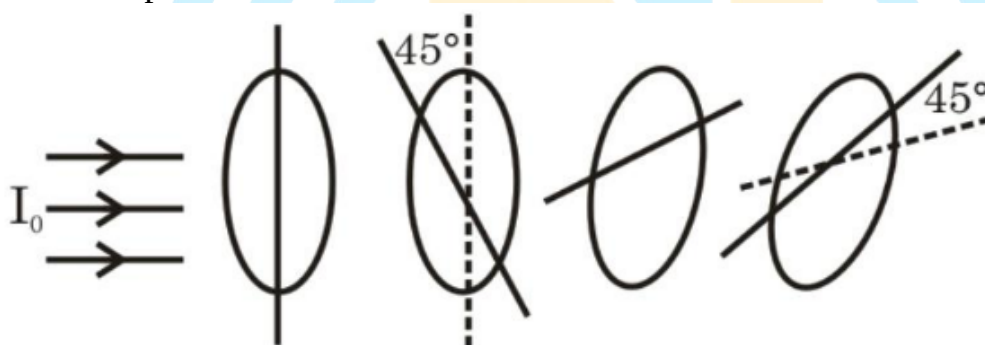
14. A 100 g of iron nail is hit by a 1.5 kg hammer striking at a velocity of 60 ms^{-1} . What will be the rise in the temperature of the nail if one fourth of energy of the hammer goes into heating the nail?
- [Specific heat capacity of iron = $0.42 \text{ Jg}^{-1}\text{C}^{-1}$]
- 1) 675°C 2) 1600°C 3) 160.7°C 4) 6.75°C
15. If the charge on a capacitor is increased by 2 C, the energy stored in it increases by 44%. The original charge on the capacitor is (in C) :
- 1) 10 2) 20 3) 30 4) 40
16. A long cylindrical volume contains a uniformly distributed charge of density ρ . The radius of cylindrical volume is R. A charge particle (q) revolves around the cylinder in a circular path. The kinetic energy of the particle is :
- 1) $\frac{\rho q R^2}{4\epsilon_0}$ 2) $\frac{\rho q R^2}{2\epsilon_0}$ 3) $\frac{q\rho}{4\epsilon_0 R^2}$ 4) $\frac{4\epsilon_0 R^2}{q\rho}$
17. An electric bulb is rated as 200 W. What will be the peak magnetic field at 4 m distance produced by the radiations coming from this bulb? Consider this bulb as a point source with 3.5% efficiency.
- 1) $1.19 \times 10^{-8} \text{ T}$ 2) $1.71 \times 10^{-8} \text{ T}$ 3) $0.84 \times 10^{-8} \text{ T}$ 4) $3.36 \times 10^{-8} \text{ T}$
18. The light of two different frequencies whose photons have energies 3.8 eV and 1.4 eV respectively, illuminate a metallic surface whose work function is 0.6 eV successively. The ratio of maximum speeds of emitted electrons for the two frequencies respectively will be :
- 1) 1 : 1 2) 2 : 1 3) 4 : 1 4) 1 : 4
19. Two light beams of intensities in the ratio of 9 : 4 are allowed to interfere. The ratio of the intensity of maxima and minima will be :
- 1) 2 : 3 2) 16 : 81 3) 25 : 169 4) 25 : 1

20. In Bohr's atomic model of hydrogen, let K, P and E are the kinetic energy, potential energy and total energy of the electron respectively. Choose the correct option when the electron undergoes transitions to a higher level :

- 1) All K, P and E increase.
- 2) K decreases. P and E increase.
- 3) P decreases. K and E increase.
- 4) K increases. P and E decrease.

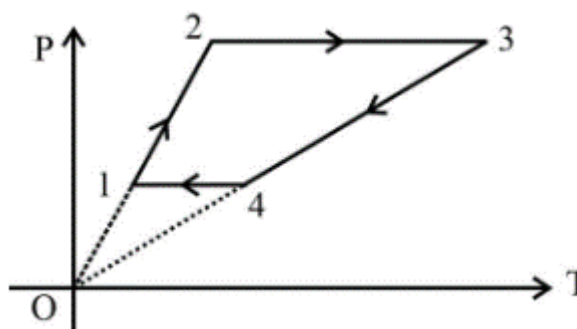
21. A body is projected from the ground at an angle of 45° with the horizontal. Its velocity after 2s is 20 ms^{-1} . The maximum height reached by the body during its motion is ____m. (use $g = 10 \text{ ms}^{-2}$)

22. Unpolarized light of intensity $I_0 = 32 \text{ w/m}^2$ is passed through four polarizers whose transmission axis are 45° w.r.t. next polarizer. What will be output intensity in w/m^2 from last polarization?



23. Three moles of an ideal monoatomic gas perform a cycle as shown in figure. The gas temperatures in different states are $T_1 = 400 \text{ K}$, $T_2 = 800 \text{ K}$, $T_3 = 2400 \text{ K}$, $T_4 = 1200 \text{ K}$. Find the work

done (in kJ) by the gas during the cycle. $\left(R = \frac{25}{3} \frac{\text{J}}{\text{mol} \cdot \text{K}}\right)$

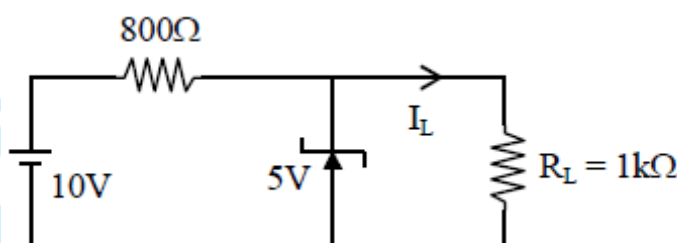


24. Two travelling waves of equal amplitudes and equal frequencies move in opposite directions along a string. They interfere to produce a stationary wave whose equation is given by

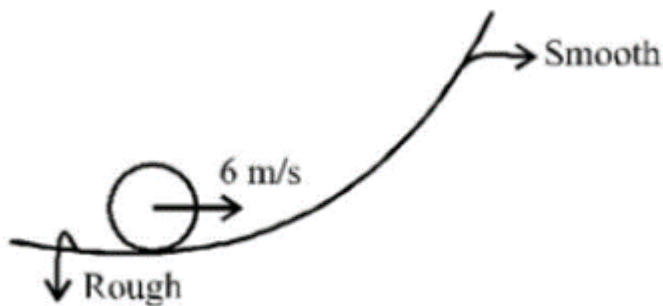
$y = 10 \cos(\pi x) \sin\left(\frac{2\pi t}{T}\right) \text{ cm}$. The amplitude of the particle at $x = \frac{4}{3} \text{ cm}$ will be ____ cm.

25. In the given circuit-the value of current I_L will be ____ mA.

(When $R_L = 1 \text{ k}\Omega$)

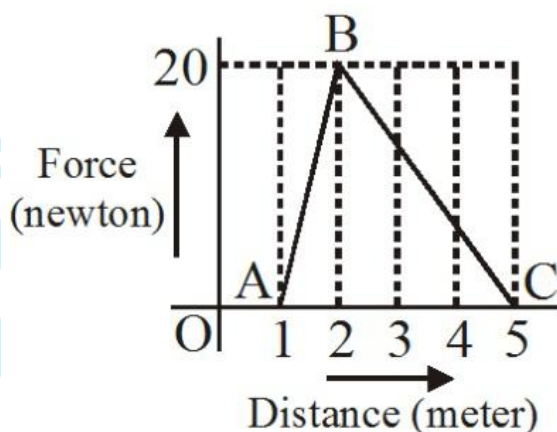


26. A disc of radius 0.1 m rolls without sliding on a horizontal surface with a velocity of 6 m/s. It then ascends a smooth continuous track as shown in figure. The height upto which it will ascend is $\frac{x}{5} \text{ m}$. Value of x is:- [Take $g = 10 \text{ m/s}^2$]



27. A ray of light is incident at an angle of incidence 60° on the glass slab of refractive index $\sqrt{3}$. After refraction, the light ray emerges out from other parallel faces and lateral shift between incident ray and emergent ray is $4\sqrt{3} \text{ cm}$. The thickness of the glass slab is ____ cm.
28. A circular coil of 1000 turns each with area 1 m^2 is rotated about its vertical diameter at the rate of one revolution per second in a uniform horizontal magnetic field of 0.07T. The maximum voltage generation will be ____ V.

29. A monoatomic gas performs a work of $\frac{Q}{4}$ where Q is the heat supplied to it. The molar heat capacity of the gas will be _____ R during this transformation. Where R is the gas constant.
30. The resisting force F acting on a body versus distance traversed by the body is plotted in figure. The mass of the body is 25 kg and the initial speed is 2 m/s. The kinetic energy (in joule) of the body at distance 2.0 m is



CHEMISTRY

MAX.MARKS: 100

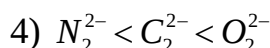
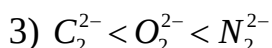
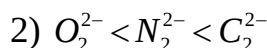
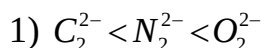
SECTION - I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

31. 120 g of an organic compound that contains only carbon and hydrogen gives 330 g of CO_2 and 270 g of water on complete combustion. The percentage of carbon and hydrogen, respectively are.
- 1) 25 and 75 2) 40 and 60 3) 60 and 40 4) 75 and 25
32. The energy of one mole of photons of radiation of wavelength 300 nm is
- (Given : $h = 6.63 \times 10^{-34} \text{ Js}$, $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$, $c = 3 \times 10^8 \text{ ms}^{-1}$)
- 1) 235 kJ mol^{-1} 2) 325 kJ mol^{-1} 3) 399 kJ mol^{-1} 4) 435 kJ mol^{-1}

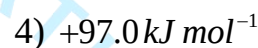
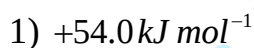
33. The correct order of bond orders of C_2^{2-} , N_2^{2-} and O_2^{2-} is, respectively.



34. At $25^\circ C$ and 1 atm pressure, the enthalpies of combustion are as given below:

Substance	H_2	C(graphite)	$C_2H_6(g)$
$\frac{\Delta_c H^\ominus}{kJ mol^{-1}}$	-286.0	-394.0	-1560.0

The enthalpy of formation of ethane is



35. For a first order reaction, the time required for completion of 90% reaction is 'x' times the half life of the reaction. The value of 'x' is

1) 1.12

2) 2.43

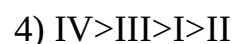
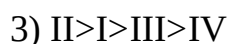
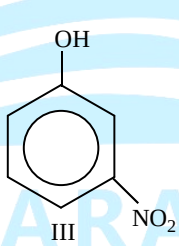
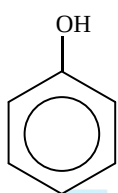
3) 3.32

4) 33.31

36. Metals generally melt at very high temperature. Amongst the following, the metal with the highest melting point will be



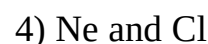
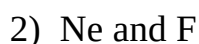
37. In the following compounds, the order of acidity is:



38. The most stable hydride is



39. The difference between electron gain enthalpies will be maximum between :



40. PCl_5 is well known, but NCl_5 is not. Because,

- 1) nitrogen is less reactive than phosphorous
- 2) nitrogen doesn't have d-orbitals in its valence shell
- 3) catenation tendency is weaker in nitrogen than phosphorous
- 4) size of phosphorous is larger than nitrogen

41. Transition metal complex with highest value of crystal field splitting (Δ_0) will be

- 1) $[Cr(H_2O)_6]^{3+}$ 2) $[Mo(H_2O)_6]^{3+}$ 3) $[Fe(H_2O)_6]^{3+}$ 4) $[Os(H_2O)_6]^{3+}$

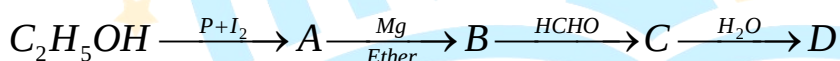
42. Assertion A : Ga exist in liquid state during summer

Reason R : Ga has unusually low melting point (303 K)

In the light of the above statements , Choose the most appropriate answer from the options given below

- 1) Both A and R are correct but R is NOT the correct explanation of A
- 2) Both A and R are correct and R is the correct explanation of A
- 3) A is not correct but R is correct
- 4) A is correct but R is not correct

43. The product in the reaction is:



- 1) propanal 2) butanal 3) n-butanol 4) n-propanol

44. Given below are two statements.

Statement I : The presence of weaker π – bonds make alkenes less stable than alkanes.

Statement II : The strength of the double bond is greater than that of carbon – carbon single bond.

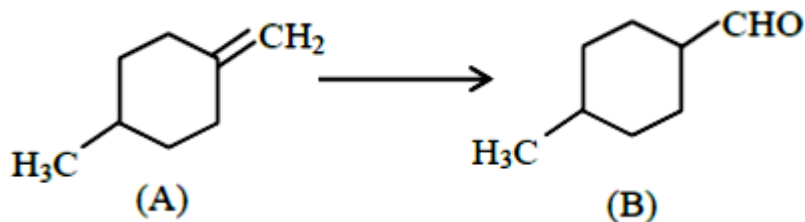
In the light of the above statements, choose the **correct** answer from the options given below.

- 1) Both Statement I and Statement II are correct
- 2) Both Statement I and Statement II are incorrect

3) Statement I is correct but Statement II is incorrect

4) Statement I is incorrect but Statement II is correct

45. Which of the following reagents / reactions will convert 'A' to 'B' ?



1) PCC oxidation

2) Ozonolysis

3) $BH_3, H_2O_2 / ^-OH$ followed by PCC oxidation

4) HBr , hydrolysis followed by oxidation by $K_2Cr_2O_7$

46. Sodium propanoate on heating with sodalime gives

1) 2 – pentene

2) propanaldehyde

3) Ethane

4) 4 – methylpent – 2 – ene

47. The conversion of propan – 1 – ol to n – butylamine involves the sequential addition of reagents. The correct sequential order of reagents is.

1) (i) $SOCl_2$ (ii) KCN (iii) $H_2 / Ni, Na(Hg) / C_2H_5OH$

2) (i) HCl (ii) $H_2 / Ni, Na(Hg) / C_2H_5OH$

3) (i) $SOCl_2$ (ii) KCN (iii) CH_3NH_2

4) (i) HCl (ii) CH_3NH_2

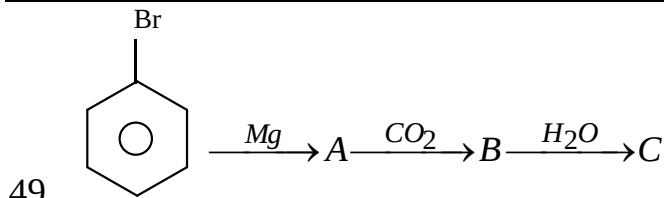
48. Fructose is a _____

1) Ketohexose

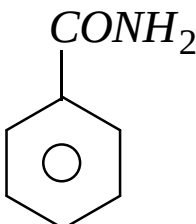
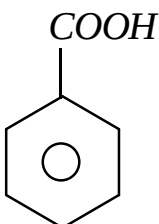
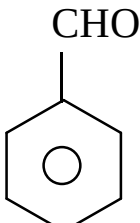
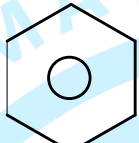
2) aldohexose

3) ketopentose

4) aldopentose



. What is C in the above conversion?

- 1)  2) 
- 3)  4) 

50. In the flame test of mixture of salts, a green flame with blue centre was observed. Which one of the following cations may be present?

- 1) Cu^{2+} 2) Sr^{2+} 3) Ba^{2+} 4) Ca^{2+}

SECTION-II (NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

51. For a mixture of two solutions, $CHCl_3$ and CH_3COCH_3 , the number of incorrect option (s) is/are :

- 1) $\Delta H < 0$
 2) $(\Delta S)_{sys} < 0$
 3) $\Delta G = -ve$
 4) Maximum boiling point azeotrope

52. How many of the following relation(s) is/are correct

- i) $\Delta G = \Delta H + T\Delta S$ ii) $\Delta G = -nFE$ iii) $\Delta H = \Delta E + \Delta nRT$ iv) $\Delta G^0 = -RT \ln K$

53. PCl_5 dissociates as $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

5 moles of PCl_5 are placed in a 200 litre vessel which contains 2 moles of N_2 and is maintained at 600 K. the equilibrium pressure is 2.46 atm. The equilibrium constant K_p for the dissociation of PCl_5 is $\text{---} \times 10^{-3}$. (nearest integer)

(Given : $R = 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1}$: assume ideal gas behaviour)

54. The resistance of conductivity cell containing 0.01 M KCl solution at 298 K is 1750Ω .

If the conductivity of 0.01 M KCl solution at 298 K is $0.152 \times 10^{-3} \text{ S cm}^{-1}$, then the cell constant of the conductivity cell is $\text{---} \times 10^{-3} \text{ cm}^{-1}$

55. How much volume (ml) of 0.1M NaOH is required for the complete neutralization of 0.1M 250ml H_2SO_4 ?

56. CO_2, SO_2, XeF_4, XeF_2 ,

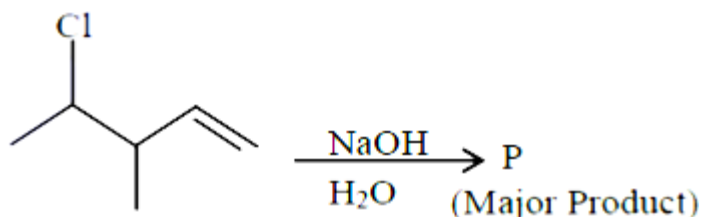
The sum of lone pairs present on central atom of each of the above molecules is ?

57. Manganese (VI) has ability to disproportionate in acidic solution. The difference in oxidation states of two ions it forms in acidic solution is ---

58. 0.2 g of an organic compound was subjected to estimation of nitrogen by Dumas method in which volume of N_2 evolved (at STP) was found to be 22.400 mL. The percentage of nitrogen in the compound is --- }nearest integer}

(Given: Molar mass of N_2 is 28 mol^{-1} . Molar volume of N_2 at STP : 22.4 L)

59.



Consider the above reaction. The number of π electrons present in the product 'P' is ____.

60. In alanylglycylleucylalanylvaline, the number of peptide linkage is ____.

MATHEMATICS**MAX.MARKS: 100**

SECTION – I
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61. Let $x * y = x^2 + y^3$ and $(x * 1) * 1 = x * (1 * 1)$. Then a value of $2\sin^{-1}\left(\frac{x^4 + x^2 - 2}{x^4 + x^2 + 2}\right)$ is
- 1) $\frac{\pi}{4}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{2}$ 4) $\frac{\pi}{6}$
62. The sum of all the real roots of the equation $(e^{2x} - 4)(6e^{2x} - 5e^x + 1) = 0$ is
- 1) $\log_e 3$ 2) $-\log_e 3$ 3) $\log_e 6$ 4) $-\log_e 6$
63. The system of equations $\alpha x + y + z = \alpha - 1, x + \alpha y + z = \alpha - 1, x + y + \alpha z = \alpha - 1$, has no solution, if α is
- 1) not -2 2) 1 3) -2 4) either -2 or 1
64. Let $x, y > 0$. If $x^3 y^2 = 2^{15}$, then the least value of $3x + 2y$ is
- 1) 30 2) 32 3) 36 4) 40
65. Which of the following is differentiable at $x=0$
- 1) $|x| + |\sin x|$ 2) $|x| - |\sin x|$ 3) $|x| + |\cos x|$ 4) $|x| - |\cos x|$

66. The value of the integral $\int_{-\pi/2}^{\pi/2} \frac{dx}{(1+e^x)(\sin^6 x + \cos^6 x)}$ is equal to
- 1) 2π 2) 0 3) π 4) $\frac{\pi}{2}$
67. If $f(n+1) = \frac{1}{2} \left(f(n) + \frac{9}{f(n)} \right)$, $n \in N$ and $f(n) > 0$ for all $n \in N$ then $\lim_{n \rightarrow \infty} f(n)$ is equal to
- 1) 3 2) $\frac{3}{2}$ 3) $\frac{1}{2}$ 4) not finite
68. Consider the points $A(0,1)$ and $B(2,0)$. P be a point on $4x+3y+9=0$ such that $|PA-PB|$ is maximum then P is
- 1) $\left(-\frac{24}{5}, \frac{17}{5}\right)$ 2) $\left(\frac{-24}{5}, \frac{-17}{5}\right)$ 3) $\left(\frac{13}{5}, \frac{-24}{5}\right)$ 4) $\left(\frac{-6}{5}, \frac{17}{5}\right)$
69. Let the maximum area of the triangle that can be inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{4} = 1$, $a > 2$, having one of its vertices at one end of the major axis of the ellipse and one of its sides parallel to the y-axis, be $6\sqrt{3}$. Then the eccentricity of the ellipse is
- 1) $\frac{\sqrt{3}}{2}$ 2) $\frac{1}{2}$ 3) $\frac{1}{\sqrt{2}}$ 4) $\frac{\sqrt{3}}{4}$
70. Let the area of the triangle with vertices, $A(1, \alpha)$, $B(\alpha, 0)$ and $C(0, \alpha)$ be 4 sq.units. If the points $(\alpha, -\alpha)$, $(-\alpha, \alpha)$ and (α^2, β) are collinear, then β is equal to
- 1) 64 2) -8 3) -64 4) 512
71. The number of distinct real roots of the equation $x^7 - 7x - 2 = 0$ is
- 1) 5 2) 7 3) 1 4) 3
72. A random variable X has the following probability distribution
- | | | | | | |
|------|---|----|----|----|----|
| X | 0 | 1 | 2 | 3 | 4 |
| P(X) | k | 2k | 4k | 6k | 8k |
- The value of $P(1 < X < 4 | X \leq 2)$ is equal to
- 1) $\frac{4}{7}$ 2) $\frac{2}{3}$ 3) $\frac{3}{7}$ 4) $\frac{4}{5}$

73. The number of solutions of the equation $\cos\left(x + \frac{\pi}{3}\right)\cos\left(\frac{\pi}{3} - x\right) = \frac{1}{4}\cos^2 2x$, $x \in [-3\pi, 3\pi]$ is
- 1) 8 2) 5 3) 6 4) 7
74. If the shortest distance between the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{\lambda}$ and $\frac{x-2}{1} = \frac{y-4}{4} = \frac{z-5}{5}$ is $\frac{1}{\sqrt{3}}$, then the sum of all possible value of λ is
- 1) 16 2) 6 3) 12 4) 15
75. The perpendicular distance of the point $P(0, -1, 3)$ from a straight line passing through $A(1, -3, 2)$ and $B(2, -1, 4)$ is
- 1) $\frac{\sqrt{29}}{3}$ 2) $3\sqrt{3}$ 3) $\frac{\sqrt{29}}{6}$ 4) $\frac{\sqrt{41}}{3}$
76. Let $\hat{a}$ and $\hat{b}$ be two units vectors such that $\left|(\hat{a} + \hat{b}) + 2(\hat{a} \times \hat{b})\right| = 2$. If $\theta \in (0, \pi)$ is the angle between $\hat{a}$ and $\hat{b}$, then among the statements
- (S1) : $2|\hat{a} \times \hat{b}| = |\hat{a} - \hat{b}|$
- (S2) The projection of $\hat{a}$ on $(\hat{a} + \hat{b}) = \frac{1}{2}$
- 1) Only (S1) is true 2) Only (S2) is true
- 3) Both (S1) and (S2) are true 4) Both (S1) and (S2) are false
77. If $K = \tan^{-1}\sqrt{\frac{a(a+b+c)}{bc}} + \tan^{-1}\sqrt{\frac{b(a+b+c)}{ca}} + \tan^{-1}\sqrt{\frac{c(a+b+c)}{ab}}$ where a, b, c are positive real numbers then $[K] = \dots$ (where $[.]$ denotes greatest integer function)
- 1) 1 2) 0 3) 3 4) 2
78. If $y = ax^{n+1} + bx^{-n}$, then $x^2 \frac{d^2y}{dx^2}$ is equal to
- 1) $n(n-1)y$ 2) $n(n+1)y$ 3) ny 4) n^2y

79. **Statement-1:** The complex number $(2+i\sqrt{5})^7 + (2-i\sqrt{5})^7$ is real

Statement -2: A complex number Z is real if and only if $\bar{Z} = Z$

1) Statement -1 is true, Statement -2 is true, Statement-2 is a correct explanation of statement-1

2) Statement-1 is true, Statement -2 is true, Statement-2 is not a correct explanation of statement-1

3) Statement -1 is true, Statement -2 is false

4) Statement -1 is false, statement -2 is true,

80. Let λ^* be the largest value of λ for which the function $f_\lambda(x) = 4\lambda x^3 - 36\lambda x^2 + 36x + 48$ is increasing for all $x \in \mathbb{R}$. Then $f_{\lambda^*}(1) + f_{\lambda^*}(-1)$ is equal to:

1) 36

2) 48

3) 64

4) 72

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

81. Let $S = \{Z \in \mathbb{C} : |z-3| \leq 1 \text{ and } z(4+3i) + \bar{z}(4-3i) \leq 24\}$. If $\alpha + i\beta$ is the point in S which is closest to $4i$, then $25(\alpha + \beta)$ is equal to.

82. Let $S = \left\{ \begin{pmatrix} -1 & a \\ 0 & b \end{pmatrix} ; a, b \in \{1, 2, 3, \dots, 100\} \right\}$ and let $T_n = \{A \in S : A^{n(n+1)} = I\}$. Then the number of elements in $\bigcap_{n=1}^{100} T_n$ is _____

83. The number of 7-digit numbers which are multiples of 11 and are formed using all the digits 1, 2, 3, 4, 5, 7 and 9 is

84. The sum of all the elements of the set $\{\alpha \in \{1, 2, \dots, 100\} : \text{HCF}(\alpha, 24) = 1\}$ is _____

85. The remainder on dividing $1 + 3 + 3^2 + 3^3 + \dots + 3^{2021}$ by 50 is _____

86. The area (in sq. units) of the region enclosed between the parabola $y^2 = 2x$ and the line $x + y = 4$ is _____.

87. Let a circle $C: (x-h)^2 + (y-k)^2 = r^2, k > 0$, touch x-axis at $(1, 0)$. If the line $x + y = 0$ intersects the circle C at P and Q such that the length of the chord PQ is 2, then the value of $h + k + r$ is equal to _____
88. In an examination, there are 10 true-false type questions. Out of 10, a student can guess the answer of 4 questions correct with probability $\frac{3}{4}$ and the remaining 6 questions correctly with probability $\frac{1}{4}$. If the probability that the student guesses the answers of exactly 8 questions correctly out of 10 is $\frac{27k}{4^{10}}$, then k is equal to _____
89. Let the hyperbola $H: \frac{x^2}{a^2} - y^2 = 1$ and the ellipse $E: 3x^2 + 4y^2 = 12$ be such that the length of latus rectum of H is equal to the length of latus rectum of E . If e_H and e_E are the eccentricities of H and E respectively, then the value of $12(e_H^2 + e_E^2)$ is equal to _____
90. If the mean and variance of eight numbers 3, 7, 9, 12, 13, 20, x and y be 10 and 25 respectively, then xy is equal to _____

